



Navigation

Agentic AI DMAIC Copilot

Home



Mode

Pilih aktivitas

- Create new project
- Open existing project



Project

New Project ID

proj_002

Start New Project



Agentic AI DMAIC Copilot - DEFINE Phase

Active Project ID: proj_002

Fase Define menggunakan Agentic AI dengan 4 kemampuan: Perception → Decision-Making → Action → Adaptation
Masukkan informasi proyek, lalu agent akan menghasilkan charter yang lengkap, SIPOC, CTQ, dan business case.



Project Metadata

Project ID

project_002

Project Name

Increase Throughput Line 1

Industry

Manufa... ▼

Process Area

Product... ▼

Primary Pain Theme

Cost ▼



Define Inputs

Problem Statement

throughput line 1 was cutrrently 5 ton/hour in jan - auguts 2025, while target is 7 ton/h with installed capacity 10 ton/h, this create cost pe rton product higher due to lower output

Goal (optional — agent will refine)

increase throughput from 5 ton/hour into 7 ton/hour

Voice of Customer (one per line)

low cost per ton

True Need / Key Issue (optional)

Upload Baseline CSV (optional)



Drag and drop file here

Limit 200MB per file • CSV

Browse files

Feedback to refine output (optional)

If result needs adjustment, enter feedback here.



Generate Draft Charter



Finalize Define Phase

Accept & Finalize

Define Phase finalized and stored in memory.

DEFINE PHASE SUMMARY — Increase Throughput Line 1

Problem Statement

Throughput line 1 was currently 5 tons/hour from January to August 2025, while the target is 7 tons/hour with an installed capacity of 10 tons/hour. This creates a higher cost per ton of product due to lower output.

Goal Statement

Increase throughput from 5 tons/hour to 7 tons/hour.

Business Case

Achieving the target throughput will reduce the cost per ton of product, enhancing profitability and operational efficiency.

Scope

- In Scope: ['Throughput line operations', 'Training programs for operators', 'Monitoring and maintenance of production equipment']
- Out of Scope: ['Other production lines not related to throughput line 1', 'Non-production related issues', 'Supplier management outside of material quality']

CTQs

- **Throughput Rate** (Tons per hour tons/hour)
- **Cost per Ton** (Cost per ton currency/ton)

- Installed Capacity Utilization (Utilization Rate %)

Self-Critique Issues (Agent Review)

```
▼ [  
  ▼ 0 : {  
    "issue" :  
      "Spelling and grammatical errors in the  
      problem text."  
    "suggestion" :  
      "Correct 'cutrrently' to 'currently' and 'pe  
      rton' to 'per ton'."  
  }  
  ▼ 1 : {  
    "issue" : "Inconsistent date format."  
    "suggestion" :  
      "Use a consistent date format throughout the  
      document, e.g., 'January to August 2025'  
      instead of 'jan - auguts 2025'."  
  }  
  ▼ 2 : {  
    "issue" :  
      "Lack of detail in the key issue field."  
    "suggestion" :  
      "Provide a specific key issue that summarizes  
      the main challenge faced in increasing  
      throughput."  
  }  
  ▼ 3 : {  
    "issue" :  
      "Missing project name in the DEFINE phase  
      summary."  
    "suggestion" :  
      "Ensure the project name is included in the  
      summary section."  
  }  
  ▼ 4 : {  
    "issue" : "Inconsistent use of terminology."
```

```
    "suggestion" :  
      "Ensure consistency in terms used for  
      metrics, e.g., 'ton/hour' vs. 'ton/h'."  
  }  
  5 : {  
    "issue" :  
      "Vague descriptions in the CTQ list."  
    "suggestion" :  
      "Provide more specific descriptions for each  
      CTQ to clarify their importance."  
  }  
  6 : {  
    "issue" :  
      "Lack of clarity in the early measure plan."  
    "suggestion" :  
      "Specify the frequency and method of  
      monitoring for each measure listed."  
  }  
]  
]
```

Final Define State JSON

```
{  
  "schema_version" : "1.0"  
  "phase" : "define"  
  "status" : "final"  
  "created_at" : "2025-12-17T08:39:41.777019+00:00"  
  "updated_at" : "2025-12-17T08:53:35.992715+00:00"  
  "project_id" : "Increase_Throughput_Line_1"  
  "inputs" : {  
    "project_name" : "Increase Throughput Line 1"  
    "industry" : "Manufacturing"  
    "process_area" : "Production"  
    "pain_theme" : "Cost"  
  }  
}
```

```
"problem_text" :
  "throughput line 1 was cutrrently 5 ton/hour
  in jan - auguts 2025, while target is 7 ton/h
  with installed capacity 10 ton/h, this create
  cost pe rton product higher due to lower
  output"
"goal_text" :
  "increase throughput from 5 ton/hour into 7
  ton/hour"
  ▼ "voc_list" : [
    |   0 : "low cost per ton"
    |
  ]
  "key_issue" : ""
}
▼ "meta" : {
  | "industry" : "Manufacturing"
  | "pain_theme" : "Cost"
  | "process_area" : "Production"
  |
}
▼ "outputs" : {
  | "schema_version" : "1.0"
  | "project_id" : "Increase_Throughput_Line_1"
  | "project_name" : "Increase Throughput Line 1"
  | "problem_statement" :
  | "Throughput line 1 was currently 5 ton/hour
  | from January to August 2025, while the target
  | is 7 ton/hour with an installed capacity of
  | 10 ton/hour. This creates a higher cost per
  | ton of product due to lower output."
  | "goal_statement" :
  | "Increase throughput from 5 ton/hour to 7
  | ton/hour."
  | "business_case" :
  | "Achieving the target throughput will reduce
  | the cost per ton of product, enhancing
  | profitability and operational efficiency."
  |
  ▼ "project_scope" : {
    |   ▼ "in_scope" : [
```

```
0 : "Throughput line operations"
1 : "Training programs for operators"
2 :
  "Monitoring and maintenance of
  production equipment"
]
▼ "out_of_scope" : [
  0 :
    "Other production lines not related
    to throughput line 1"
  1 : "Non-production related issues"
  2 :
    "Supplier management outside of
    material quality"
]
}
▼ "ctq_list" : [
  ▼ 0 : {
    "name" : "Throughput Rate"
    "description" :
      "The amount of product produced per
      hour on the throughput line, aiming
      for a target of 7 ton/hour."
    "metric" : "Tons per hour"
    "unit" : "ton/h"
  }
  ▼ 1 : {
    "name" : "Cost per Ton"
    "description" :
      "The total cost incurred to produce
      one ton of product, aiming to reduce
      costs through increased throughput."
    "metric" : "Cost per ton"
    "unit" : "currency/ton"
  }
  ▼ 2 : {
    "name" :
      "Installed Capacity Utilization"
```

```
"description" :  
  "The percentage of the installed  
  capacity that is being utilized for  
  production, aiming for optimal  
  utilization."  
  "metric" : "Utilization Rate"  
  "unit" : "%"  
}  
]  
"y_variable" : "Throughput Rate"  
▼ "x_categories" : {  
  ▼ "Man" : [  
    0 :  
      "Operator training on production  
      processes"  
    1 :  
      "Staffing levels on throughput line"  
  ]  
  ▼ "Machine" : [  
    0 :  
      "Condition and maintenance of  
      production equipment"  
    1 :  
      "Calibration of machinery for optimal  
      performance"  
  ]  
  ▼ "Method" : [  
    0 :  
      "Production process standard  
      operating procedures"  
    1 :  
      "Efficiency of workflow and task  
      allocation"  
  ]  
  ▼ "Material" : [  
    0 :  
      "Quality of raw materials used in  
      production"
```



```
1 :  
  "Supplier consistency and  
  reliability"  
]  
▼ "Measurement" : [  
  0 :  
    "Monitoring throughput rates  
    regularly"  
  1 :  
    "Data collection methods for  
    production output"  
]  
▼ "Environment" : [  
  0 :  
    "Production area layout and  
    ergonomics"  
  1 :  
    "Temperature and humidity control in  
    the production area"  
]  
}  
▼ "sipoc" : {  
  ▼ "Suppliers" : [  
    0 : "Raw material suppliers"  
    1 : "Equipment manufacturers"  
  ]  
  ▼ "Inputs" : [  
    0 : "Raw materials"  
    1 : "Production equipment"  
    2 : "Operator training"  
  ]  
  ▼ "Process" : [  
    0 : "Production process"  
    1 : "Quality control checks"  
  ]  
  ▼ "Outputs" : [  
    0 : "Finished products"  
  ]  
}
```

```
  ▼ "Customers" : [  
    0 : "End customers"  
    1 : "Distributors"  
  ]  
}  
  ▼ "early_measure_plan" : {  
    ▼ "measures" : [  
      0 : "Monitor throughput rates weekly"  
      1 : "Track cost per ton bi-weekly"  
      2 :  
        "Conduct regular maintenance checks  
        on production equipment"  
    ]  
    "responsible" :  
      "Production Management Team"  
  }  
  "risk_level" : "Medium"  
  "project_type" : "Process Improvement"  
}  
"insight_md" : "" 
```

```
"summary_md" :  
"## DEFINE PHASE SUMMARY – Increase Throughput  
Line 1  
  
### Problem Statement  
Throughput line 1 was currently 5 tons/hour from  
January to August 2025, while the target is 7  
tons/hour with an installed capacity of 10  
tons/hour. This creates a higher cost per ton of  
product due to lower output.  
  
### Goal Statement  
Increase throughput from 5 tons/hour to 7  
tons/hour.  
  
### Business Case  
Achieving the target throughput will reduce the  
cost per ton of product, enhancing profitability  
and operational efficiency.  
  
### Scope  
- In Scope: ['Throughput line operations',  
'Training programs for operators', 'Monitoring  
and maintenance of production equipment']  
- Out of Scope: ['Other production lines not  
related to throughput line 1', 'Non-production  
related issues', 'Supplier management outside of  
material quality']  
  
### CTQs  
- **Throughput Rate** (Tons per hour tons/hour)  
- **Cost per Ton** (Cost per ton currency/ton)  
- **Installed Capacity Utilization** (Utilization  
Rate %)"  
  
"project_name" : "Increase Throughput Line 1"  
  
"problem_statement" :  
"Throughput line 1 was currently 5 tons/hour from  
January to August 2025, while the target is 7  
tons/hour with an installed capacity of 10  
tons/hour. This creates a higher cost per ton of  
product due to lower output."
```

```
"goal_statement" :
"Increase throughput from 5 tons/hour to 7
tons/hour."

"business_case" :
"Achieving the target throughput will reduce the
cost per ton of product, enhancing profitability
and operational efficiency."

▼ "project_scope" : {
  ▼ "in_scope" : [
    0 : "Throughput line operations"
    1 : "Training programs for operators"
    2 :
      "Monitoring and maintenance of production
      equipment"
  ]
  ▼ "out_of_scope" : [
    0 :
      "Other production lines not related to
      throughput line 1"
    1 : "Non-production related issues"
    2 :
      "Supplier management outside of material
      quality"
  ]
}

▼ "ctq_list" : [
  ▼ 0 : {
    "name" : "Throughput Rate"
    "description" :
      "The amount of product produced per hour
      on the throughput line, aiming for a
      target of 7 tons/hour."
    "metric" : "Tons per hour"
    "unit" : "tons/hour"
  }
  ▼ 1 : {
    "name" : "Cost per Ton"
```

```
"description" :  
  "The total cost incurred to produce one  
  ton of product, aiming to reduce costs  
  through increased throughput."  
  "metric" : "Cost per ton"  
  "unit" : "currency/ton"  
}  
▼ 2 : {  
  "name" : "Installed Capacity Utilization"  
  "description" :  
    "The percentage of the installed capacity  
    that is being utilized for production,  
    aiming for optimal utilization."  
  "metric" : "Utilization Rate"  
  "unit" : "%"  
}  
]  
"y_variable" : "Throughput Rate"  
▼ "x_categories" : {  
  ▼ "Man" : [  
    0 :  
      "Operator training on production  
      processes"  
    1 : "Staffing levels on throughput line"  
  ]  
  ▼ "Machine" : [  
    0 :  
      "Condition and maintenance of production  
      equipment"  
    1 :  
      "Calibration of machinery for optimal  
      performance"  
  ]  
  ▼ "Method" : [  
    0 :  
      "Production process standard operating  
      procedures"
```

```
1 :  
  "Efficiency of workflow and task  
  allocation"  
]  
▼ "Material" : [  
  0 :  
    "Quality of raw materials used in  
    production"  
  1 : "Supplier consistency and reliability"  
]  
▼ "Measurement" : [  
  0 :  
    "Monitoring throughput rates regularly"  
  1 :  
    "Data collection methods for production  
    output"  
]  
▼ "Environment" : [  
  0 :  
    "Production area layout and ergonomics"  
  1 :  
    "Temperature and humidity control in the  
    production area"  
]  
}  
▼ "sipoc" : {  
  ▼ "Suppliers" : [  
    0 : "Raw material suppliers"  
    1 : "Equipment manufacturers"  
  ]  
  ▼ "Inputs" : [  
    0 : "Raw materials"  
    1 : "Production equipment"  
    2 : "Operator training"  
  ]  
  ▼ "Process" : [  
    0 : "Production process"  
    1 : "Quality control checks"
```

```
]
  "Outputs" : [
    0 : "Finished products"
  ]
  "Customers" : [
    0 : "End customers"
    1 : "Distributors"
  ]
}
"early_measure_plan" : {
  "measures" : [
    0 : "Monitor throughput rates weekly"
    1 : "Track cost per ton bi-weekly"
    2 :
      "Conduct regular maintenance checks on
      production equipment"
  ]
  "responsible" : "Production Management Team"
}
"risk_level" : "Medium"
"project_type" : "Process Improvement"
}
```



Export Define Report

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